

Mohammed Mahdi Nsr

AI Engineer

✉ mohammedmahdi.ai@gmail.com ☎ +201028208558 📍 Egypt-Cairo 🌐 LinkedIn 🐙 Github

Summary

AI Engineer focused on developing intelligent AI systems powered by LLMs, RAG, and Multi-Agent architectures, with expertise in Machine Learning, Deep Learning, NLP and Computer vision. Experienced in transforming cutting-edge AI research into scalable, production-ready solutions using LangChain, LangGraph, CrewAI, Hugging Face, and vector databases

Education

Faculty of computers and Artificial intelligence 2020 – 2024 | Beni Suef university
Bachelor's degree in Computer Science
Grade : Very Good

EXPERIENCE

IT Specialist & System Administrator — Military Service 01/2025 – 03/2026

Prodigy InfoTech 07/2024 – 09/2024

Designed and implemented two end-to-end machine learning projects as part of AI & Data Science training:

Built two end-to-end ML projects:

- Movie Recommender System (Content-based filtering for movie suggestions)
- Sentiment Analysis on Amazon Reviews (NLP classification of customer feedback)

ITI Summer Trainig, AI & Data Science 08/2023 – 09/2023

Completed intensive training program in Artificial Intelligence and Data Science at Information Technology Institute (ITI).

Gained strong understanding of core AI concepts including machine learning fundamentals, data preprocessing, model evaluation, and basic neural networks.

Developed hands-on experience in applying AI techniques to real-world problems.

Built two end-to-end projects demonstrating practical implementation of learned concepts.

Projects

1 - Egyptian-Legal-Research-Agent , 05/2026 – 05/2026

Multi-Agent Legal Research AI System for Egyptian Civil Law

- Built a Multi-Agent RAG-based Legal Research System for Egyptian Civil Law using LangGraph with specialized agents for Orchestration, Retrieval, Web Search, Reranking, Critique, and Answer Generation.
- Implemented semantic retrieval and cross-encoder reranking pipelines to improve document relevance and answer quality.
- Integrated multiple LLM providers to support flexible model routing and optimized inference across the research workflow.
- Reduced hallucinations by grounding responses in retrieved legal documents, augmenting knowledge with web search, reranking retrieved evidence, and validating outputs through a dedicated Critic Agent before final answer generation.
- Designed an end-to-end multi-agent workflow that coordinates retrieval, verification, reasoning, and answer generation to deliver reliable legal research assistance.
- Containerized and deployed the application using Docker and FastAPI.

Tech Stack: Python, LangGraph, LangChain, RAG, Multi-Agent Systems, Google Gemini, Groq, Tavily Search, FastAPI, Docker, intfloat/multilingual-e5-small, cross-encoder/mmarco-mMiniLMv2-L12-H384-v1.

2 - LLM Fine-Tuning for Docker Command Generation 04/2026

- Fine-tuned Hugging Face Qwen2.5-Coder-1.5B-Instruct using LoRA (Low-Rank Adaptation) to generate Docker CLI commands from natural language instructions.
- Implemented supervised fine-tuning (SFT) workflows using LlamaFactory, Docker, and Linux-based environments.
- Built an instruction-following NLP system capable of converting user requests into valid Docker commands with optimized inference efficiency.
- Evaluated and tested the fine-tuned model using custom prompting and response generation pipelines.

3 - Multilingual Text Classification System , 03/2026 – 04/2026

Designed and implemented an end-to-end multilingual NLP system for detecting and classifying Arabic and English text using machine learning techniques.

- **1. Arabic Text Classification** Built an Arabic text classification pipeline including data preprocessing, EDA, TF-IDF feature extraction, model training, and evaluation. Performed exploratory data analysis covering class distribution, text length analysis, frequent word extraction, and word cloud visualization. Trained and evaluated the model using Gaussian Naïve Bayes with classification reports and confusion matrices.
- **2. English Text Classification** Developed a machine learning model to automatically distinguish between Arabic and English text. Combined Arabic and English datasets to improve generalization and overall model performance.
- **3. Full System Implementation** Implemented the final solution as a structured Python project with a modular architecture, including separate components for: Language Detection / Arabic Text Classification / English Text Classification / Main Application Workflow

Skills

Programming Principles (• Problem solving | • OOP | • Database (Sql server) | • Data structures | • Algorithms | • Math for AI | • Python)

AI & ML Concepts (• EDA | • Data Preprocessing | • Text Preprocessing | • NLTK & Spacy | • Transformers | • Python AI libraries | • ML algorithms | • Deep learning | • AI frameworks)

Agentic AI & LLMs (• Agentic AI & LLMs | • AI Agents | • GenAI | • HuggingFace | • LangChain | • LangGrapah | • CrewAI | • RAG | • Fine-tuning | • VLMs (OCR) | • Vector database (Chroma, Faiss & Pinecone))

Deployment (• APIs (FastAPIs) | • Git & GitHub | • Docker | • AWS | • Google Cloud Platform (GCP))

Languages

Arabic — Native/Bilingual

English — Proficient